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Shifted Moving Block Bootstrap for Granger Causality

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Background

- Granger causality is a measure of unique predictive information passed from one variable to another, and is particularly prevalent in econometrics (Granger, 1969; Shojaie and Fox, 2021). Researchers often use Granger causality tests for variable selection in forecasting.
- For data generated by models outside of the Gaussian $\text{VAR}(p)$ framework, non-parametric resampling methods are often preferred to distribution-based classical tests (Lütkepohl, 2005).
- We propose a novel bootstrap procedure to test a null hypothesis of Granger non-causality in a bivariate setting, adapted from a test introduced by Yin et al. (2025) for extremal compound structural causality.

Test Motivation

- A consistent bootstrap procedure must reconstruct the null hypothesis of non-causality, so we must retain within-series autocorrelation, while breaking cross-series causation to create an empirical null space.
- Our chosen bootstrap statistic is the F -statistic from Granger's classical F -test (Granger, 1969). This is a measure with rich literature providing existing results for us to use in proving the asymptotic behaviour of our test.
- We draw on existing literature for M -estimators to provide two bias corrections to a naive bootstrap approach (Lahiri, 2003; Hall and Horowitz, 1996).

Test Overview

Consider (X, Y) to be a realisation of length n of some bivariate stochastic process indexed by $\mathbb{Z}$.

Let the presence (and direction) of Granger causality from the model generating X and Y be denoted as $X \rightarrow Y$, with Granger non-causality denoted as $X \nrightarrow Y$. We can then consider a set of hypotheses for our procedure to test to some significance level $\alpha \in (0, 1)$:

$$H_0 : X \nrightarrow Y, \quad H_1 : X \rightarrow Y.$$

Our test statistic is of the form:

$$F_{X \rightarrow Y} = \frac{n - 3p - 1}{p} \left(\frac{\text{RSS}_0(X, Y) - \text{RSS}_1(X, Y)}{\text{RSS}_1(X, Y)} \right),$$

for some estimated autoregressive lag p .

Test Overview - Step 1

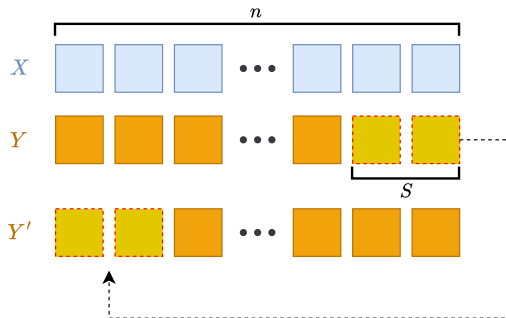
We firstly construct a shifted series:

$$Y'_t = Y_{(t-S) \bmod n}.$$

If we take S to be growing unbounded with n , for an α -mixing series we will have:

$$\text{Cov}(X_t, Y'_t) \rightarrow 0, \quad \forall t \in \{1, \dots, n\}.$$

This acts to enforce the null of Granger non-causality; however, it is bi-directional, which is stronger than H_0 .

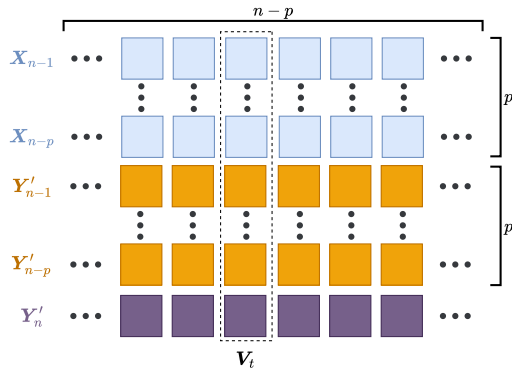


Test Overview - Step 2

Using an estimated maximum autoregressive lag p , we construct the following regression segments necessary for our bootstrap statistic:

$$\mathbf{V}_t = \begin{pmatrix} \mathbf{X}_t \\ \mathbf{Y}'_t \\ Y'_{t+1} \end{pmatrix},$$

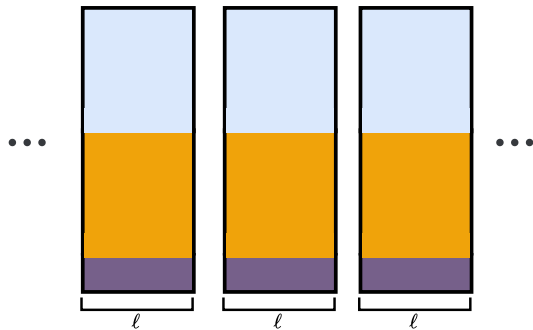
$$\mathbf{X}_t = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-p+1} \end{pmatrix}, \quad \mathbf{Y}'_t = \begin{pmatrix} Y'_t \\ Y'_{t-1} \\ \vdots \\ Y'_{t-p+1} \end{pmatrix}.$$



Test Overview - Step 3

We then resample blocks of length ℓ from $\{\mathbf{V}_p, \dots, \mathbf{V}_{n-1}\}$ using the overlapping moving block bootstrap.

On this concatenated series, we can calculate a bootstrap statistic, and we can make a decision on whether to reject the null hypothesis based on the proportion of bootstrap repetitions in which the bootstrap statistic is larger than $F_{X \rightarrow Y}$.



Adjusted Bootstrap Statistic - Motivation and Construction

In general, using the F -statistic as is will cause our test to be conservative, that is:

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\text{Reject } H_0 \mid H_0) < \alpha.$$

While this strict inequality is advantageous in the asymptotic case, it will cause a power trade-off in the finite sample case. We therefore propose an adjustment to the procedure for the finite sample case, whereby we adjust the regression segments as follows:

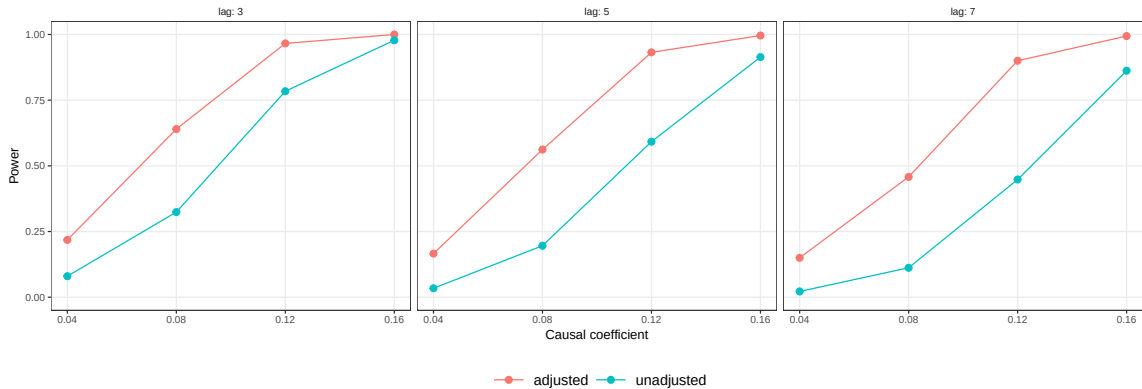
$$\mathbf{V}_t = \begin{pmatrix} \mathbf{X}_t \\ \mathbf{Y}'_t \\ \tilde{\mathbf{Y}}'_{t+1} \end{pmatrix},$$

where:

$$\tilde{\mathbf{Y}}'_{t+1} = \mathbf{Y}'_{t+1} - \mathbf{X}_t^\top \hat{\boldsymbol{\delta}},$$

for an estimated causal coefficient matrix $\hat{\boldsymbol{\delta}}$.

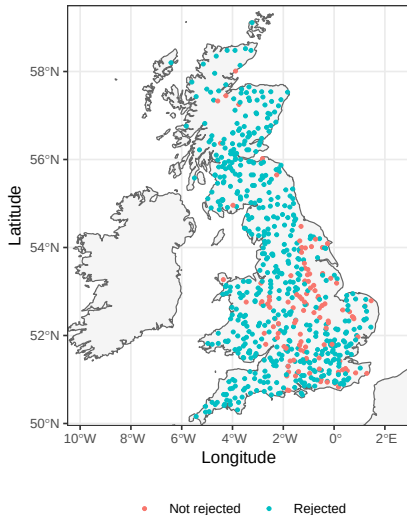
Adjusted Bootstrap Statistic - Implementation



Application to UK River Flow and Precipitation Data

We apply our procedure to a real-world dataset. CAMELS-GB is a dataset of daily readings of ecological variables across UK sites from 30/09/1970 to 30/09/2015 (Coxon et al., 2020). We use the river flow and precipitation data to test our procedure.

The null hypothesis of precipitation not Granger causing river flow was rejected for every station, while the reverse null hypothesis (river flow does not Granger cause precipitation) was partly rejected, with a clear spatial clustering governing these rejections.



Summary - Theoretical Results

Our report introduces a novel bivariate Granger causality test, and we prove the following:

- For α -mixing processes (under weak moment conditions), the procedure is asymptotic pointwise consistent, that is:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\text{Reject } H_0 \mid H_1) = 1.$$

- For a VAR(p) process with Gaussian errors and with the adjusted F -statistic, the procedure is asymptotic pointwise exact at α , that is:

$$\limsup_{n \rightarrow \infty} \mathbb{P}(\text{Reject } H_0 \mid H_0) = \alpha.$$

Summary - Extensions

Directions for future research include:

- General multivariate case – accounting for confounding variables and non-stationary series.
- Test statistic – is the F -statistic limiting the gains made by this procedure over classical tests?
- Extending theory – can we provide theoretical results on the asymptotic behaviour under the null for general α -mixing models? Can we provide a rate of convergence of the shifted series to the null distribution?

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Thank you.

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References I

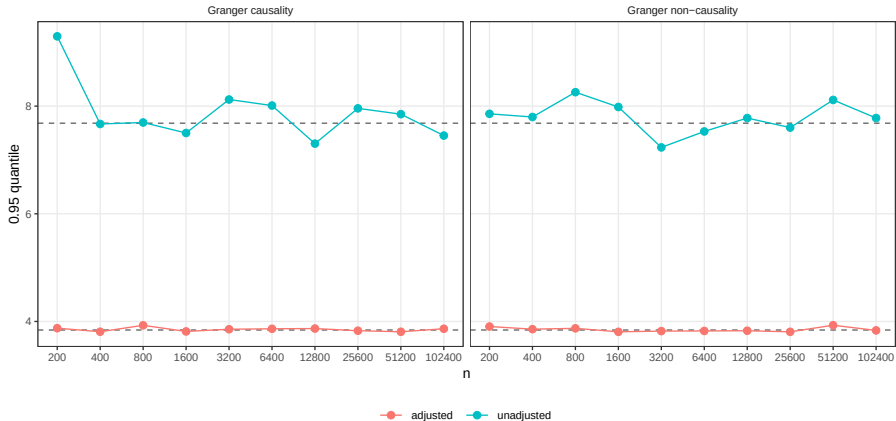
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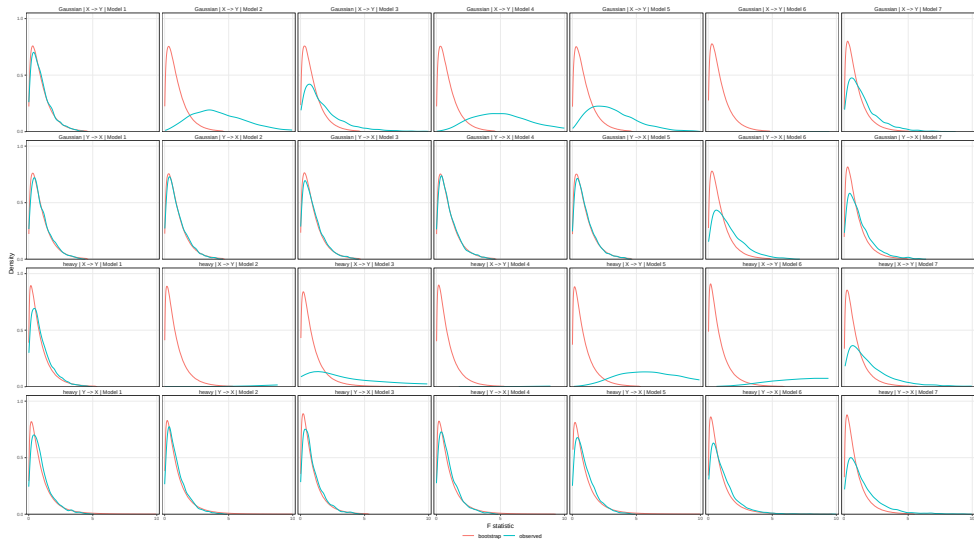
Convergence of F -statistics



Monte Carlo Simulations - Models

Model	Structure	Causal mechanism	Causal direction
1	White noise	None	None
2	Marginal white noise	Linear	$X \rightarrow Y$
3	Marginal white noise	Non-linear	$X \rightarrow Y$
4	Vector autoregressive	Linear	$X \rightarrow Y$
5	Moving average	Non-linear	$X \rightarrow Y$
6	GARCH	Linear	$X \rightarrow Y$
7	GARCH	Variance	$X \rightarrow Y$

Monte Carlo Simulations - F -statistic distribution



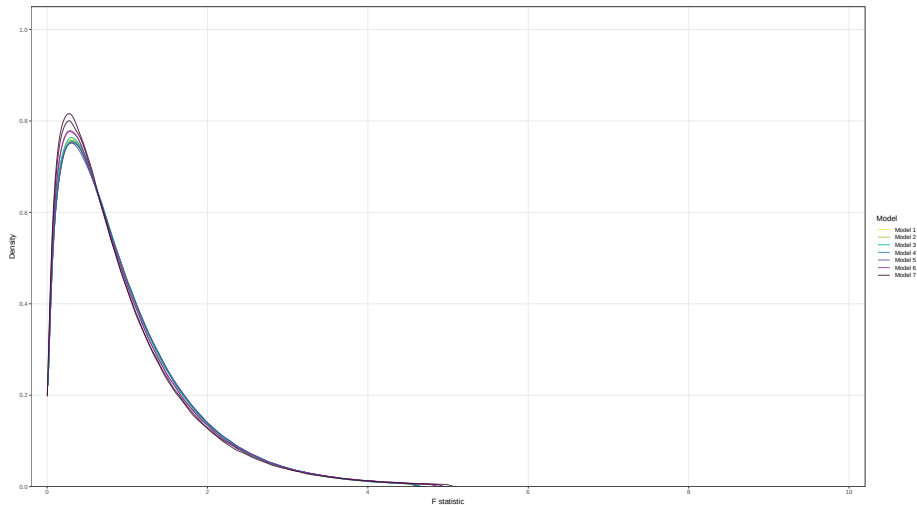
Monte Carlo Simulations - p -values for Gaussian models

Model	SMBB				Existing tests			
	1	2	3	4	F -test	R.Wald	AR Sieve	TY
$X \rightarrow Y$								
1	0.0504	0.0104	0.0496	0.0120	0.0490	0.0514	0.0490	0.0490
2	0.7512	0.4262	0.7508	0.4572	0.7602	0.7574	0.7586	0.7570
3	0.2298	0.1002	0.2256	0.1052	0.2296	0.0518	0.2228	0.2282
4	0.8638	0.5748	0.8618	0.6142	0.8700	0.8690	0.8678	0.7628
5	0.5422	0.2524	0.5386	0.2764	0.5532	0.5460	0.5450	0.6208
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
7	0.1730	0.0632	0.1674	0.0716	0.1648	0.0546	0.1618	0.1544
$Y \rightarrow X$								
1	0.0494	0.0066	0.0476	0.0090	0.0454	0.0468	0.0468	0.0462
2	0.0512	0.0098	0.0468	0.0092	0.0450	0.0452	0.0472	0.0458
3	0.0562	0.0100	0.0540	0.0126	0.0490	0.0520	0.0506	0.0504
4	0.0554	0.0080	0.0526	0.0092	0.0508	0.0538	0.0512	0.0532
5	0.0552	0.0096	0.0538	0.0124	0.0498	0.0514	0.0490	0.0502
6	0.1860	0.0562	0.1794	0.0638	0.1848	0.1360	0.1744	0.0998
7	0.1200	0.0372	0.1160	0.0408	0.1002	0.0538	0.0966	0.1034

Monte Carlo Simulations - p -values for heavy tailed models

Model	SMBB				Existing tests			
	1	2	3	4	F -test	R.Wald	AR Sieve	TY
$X \rightarrow Y$								
1	0.0752	0.0194	0.0696	0.0208	0.0576	0.0752	0.0550	0.0566
2	1.0000	0.9964	1.0000	0.9974	1.0000	0.9988	1.0000	1.0000
3	0.5902	0.4674	0.5876	0.4778	0.5960	0.0420	0.5766	0.5946
4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0.9268	0.7200	0.9290	0.7454	0.9390	0.8842	0.9350	0.9618
6	0.9824	0.9016	0.9826	0.9190	0.9890	0.9660	0.9882	0.9916
7	0.2964	0.1564	0.2846	0.1632	0.2838	0.0544	0.2714	0.2658
$Y \rightarrow X$								
1	0.0552	0.0184	0.0532	0.0202	0.0566	0.0404	0.0554	0.0574
2	0.0470	0.0184	0.0464	0.0200	0.0462	0.0390	0.0468	0.0452
3	0.0694	0.0218	0.0648	0.0218	0.0508	0.0536	0.0470	0.0512
4	0.0516	0.0184	0.0474	0.0192	0.0518	0.0374	0.0514	0.0510
5	0.0516	0.0192	0.0524	0.0222	0.0558	0.0458	0.0502	0.0536
6	0.0878	0.0448	0.0834	0.0482	0.1002	0.0526	0.0906	0.0774
7	0.1850	0.0802	0.1766	0.0870	0.1620	0.0510	0.1534	0.1608

$\tilde{F}_{X \rightarrow Y'}^*$ distribution over different Gaussian models



$\tilde{F}_{X \rightarrow Y'}^*$ distribution over different heavy-tailed models

